TECHMASTERS INNOVATIONS PRIVATE LIMITED

BAGODI LUXURY E-COMMERCE

Physical Shopping Mall & Multi-Brand Fashion Ecosystem

FLAGSHIP STORE: SHIVAJI CHOWK, BIDAR, KARNATAKA | RC v1.0.0

1. EXECUTIVE SUMMARY & ARCHITECTURAL SCOPE

The Bagodi Digital Ecosystem is an enterprise-grade luxury retail platform engineered as the direct digital extension of the physical Bagodi Shopping Mall in Bidar, Karnataka. Featuring premier national partner brands including Raymond, Park Avenue, Parx, and ColorPlus, alongside Bagodi Couture Silk Sarees and Bridal Lehengas, the architecture guarantees an internationally competitive shopping experience while retaining authentic Indian retail heritage.

2. CORE ARCHITECTURAL PILLARS

A. Single Codebase Mobile Shell
One customer codebase powers both Web (:3000) and Native Android APK via Capacitor 6.

B. Server-Authoritative Integrity
Zero client price computations. Totals, GST, and coupons verified via FastAPI contracts.

C. Physical Mall Barcode HUD
USB/Bluetooth scanner listener with 60ms debouncing for instant stock intake at Bidar.

D. Pure Plain CSS Token System
Zero Tailwind/Bootstrap. Strict CSS modules with 3D tactile perspective and Obsidian/Gold palette.

E. Multi-Tenant Governance
Decoupled Super Admin portal (:3002) for license issuance, RBAC, and immutable audit trails.

AUDIT VERIFICATION STATUS: **10 / 10 PHASES FULLY IMPLEMENTED & OPERATIONAL**

3. PHASE DELIVERY AUDIT: FOUNDATION & STOREFRONT (PHASES 0 - 4)

PHASE	MODULE NAME	DELIVERABLES & TECHNICAL SCOPE	STATUS
Phase 0	Monorepo & Foundation	npm workspaces, Plain CSS tokens, 3D perspective elevation (1000px), shared UI component library, Axios client with JWT interceptors.	VERIFIED
Phase 1	Authentication & RBAC	Customer phone OTP + password flows, Merchant staff credentials, Super Admin master session, permission guards, hybrid token storage adapter.	VERIFIED
Phase 2	Catalogue & Variants	Category navigation tree, faceted server filters (Brand, Fabric, Size, Color), PDP gallery, variant SKU resolution, product manager modal.	VERIFIED
Phase 3	18-Screen Inventory HUD	Stock ledger, rapid stock adjustment workbench, USB/Bluetooth barcode HUD modal, batch intake, low-stock threshold monitors.	VERIFIED
Phase 4	Shopping & Checkout	Server-authoritative cart, multi-address manager, Razorpay payment ga n, order confirmation lifecycle, privilege coupons.	VERIFIED

RAYMOND AUTHORISED PARTNERSHIP INTEGRATION:

Full bespoke suiting, blazers, and unstitched fabric cataloguing with custom tailoring notes

4. PHASE DELIVERY AUDIT: REDESIGN, LOGISTICS & GOVERNANCE (PHASES 5 - 9)

PHASE	MODULE NAME	DELIVERABLES & TECHNICAL SCOPE	STATUS
Master UI	**Editorial Luxury Redesign**	Warm Ivory (#F7F1E7) background, Cormorant Garamond typography, [illegible]shmiri couture cinematic animated hero, multi-layer header.	VERIFIED
Phase 5	**Customer Android App**	Capacitor 6 native shell, hardware back-button listener, status bar styling (#202321), physical haptics, deep linking (bagodi://), safe-area insets.	VERIFIED
Phase 6	**Merchant Order Admin**	Order queue (CONFIRMED -> PACKED -> SHIPPED), printable 80mm ESC/POS packing slips, return/exchange pass/fail inspection workbench.	VERIFIED
Phase 7	**Logistics & Analytics**	Multi-carrier courier fleet (Shiprocket, Delhivery, BlueDart), driver handov[illegible] dynamic Privilege Coupon engine, margin reporting.	VERIFIED
Phase 8	**Super Admin Governance**	Platform telemetry, multi-tenant merchant onboarding, operator RBAC a[illegible]utable security audit trail, emergency maintenance dials.	VERIFIED
Phase 9	**Production Readiness**	Rollup chunk code-splitting (<200 kB), offline network resilience sentinel[illegible] multi-stage Dockerfiles, Nginx reverse proxy, CI/CD pipeline.	VERIFIED

CODE QUALITY & VERIFICATION SIGN-OFF:

Monorepo TypeScript check: 0 errors | Vite production builds: 0 errors | All portals tested

5. LUXURY DESIGN SYSTEM & TOKEN SPECIFICATIONS

The design system enforces a zero-utility-CSS rule with pure CSS custom properties and CSS Modules:

Obsidian Charcoal `#202321`	Dark surfaces, Admin shells, Top customer navigation bar
Heritage Olive `#34392D`	Secondary dark surfaces, architectural building textures
Champagne Gold `#C5A15B`	Primary metallic luxury accent, badges, dividers, borders
Warm Ivory `#F7F1E7`	Editorial light customer background, saree showcase backdrop
Royal Burgundy `#641F32`	Festive fashion campaigns, bridal lehengas, hero highlights
Heritage Red `#B5262E`	Brand logo accent, controlled sale tags, promotional pills
Pure White `#FFFFFF`	Product card tiles for maximum garment fabric color fidelity

6. TYPOGRAPHY & HIERARCHY SPECIFICATIONS

Display Headings:	**Playfair Display / Cormorant Garamond**	Editorial hero titles, DAR-E-AAB campaign, brand storytelling
Interface Body:	**Inter / Manrope**	Product titles, catalog prices, specifications, checkout forms
Telemetry & Ledger:	**JetBrains Mono / System Monospace**	Barcode SKU codes, courier AWB tracking, inventory audit entries

7. MULTI-PORTAL DEPLOYMENT TOPOLOGY & PORT REFERENCE

PORTAL	PORT	HOSTNAME	CAPABILITIES & TARGET AUDIENCE
Customer Web	`:3000`	bagodi.in	Editorial luxury storefront, shopping cart, Razorpay checkout
Merchant Admin	`:3001`	admin.bagodi.in	18-screen inventory cockpit, barcode HUD, returns workbench
Super Admin	`:3002`	super.bagodi.in	Platform governance, multi-tenant onboarding, audit log ledger
Customer Android	APK	Google Play Store	Capacitor 6 native shell, hardware back button, haptics
Gateway Nginx	`:80 / :443`	Reverse Proxy	HTTP/2, Gzip/Brotli, SSL termination, CSP security headers

8. PHYSICAL STORE HARDWARE ONBOARDING (BIDAR MALL)

1. USB Barcode Scanners (Zebra LS2208 / Honeywell): Configured for HID keyboard emulation with [ENTER] suffix. Automatic 60ms sequence detection opens the Barcode HUD modal in <100ms.

2. Thermal Receipt Printers (Epson / TVS RP3200): Pre-configured @media print CSS stylesheets ensure crisp 80mm dispatch manifests and store pickup receipts without dialog formatting issues.

3. Offline Resilience: Built-in NetworkStatusBar automatically buffers user state and notifies staff during physical internet drops, auto-syncing upon cloud reconnection.

EXECUTIVE ENGINEERING SIGN-OFF

PLATFORM: Bagodi Luxury Fashion & Multi-Brand E-Commerce Platform

DEVELOPMENT STATUS: Complete (Phases 0 through 9 + Master UI Redesign Verified)

SIGNED: Senior Luxury Frontend Architect | TechMasters Innovations Private Limited